Java is OO, widely used, powerful, flexible, portable, secure not the fastest language. Needs one or more class definitions.
OO is cool because objects can be used efficiently to represent real-world entities. Principles of OO are objects, attribute, method, class, encapsulation, inheritace, polymorphism.
INTERPRETER – runs the program
COMPLIER – needed for creating programs
DATA FIELD – named piece of data
METHODS – contain statements (steps) of programs
OBJECTS – instances of classes
ALGORITHM – description of processes
PRIMTIVE TYPE – float,long,byte,short,int,Boolean,char
REFERENCE TYPE – references to instance objects
VARIABLE – named bit of data. Declared & initialised
CLASS OBJECTS – one created automatically
INSTANCE OBJECTS - any number can be created (eg. Box box1 = new Box(x);)
APPLICATION – program 1 or more classes. One must be the application class (with main)
CLASS – declaration/header. Data fileds, methods ≤0
METHOD – decl/head. Local var, other statements ≤0
DATA FIELDS – used everywhere, specified visibility (need to be static if in class object), may be initialised
LOCAL VARIABLES – only used in their method, no need for visibility/static, must be initialised
JFC – java foundation classes (libraries, class libs, API)
API – application programming interface
GUI – Graphical User Interface
IDE – integrated development environment
AWT – abstract windowing toolkit
UMI – Unified Modelling Language
SUPPORT CLASS – non-static data fields and methods, instance objects made from these
APPLICATION CLASS – static data fields and methods
ENCAPSULATED – data protected by methods.
ACCESSOR – return the value in a data field. empty()
public String getName() { return name; }
MUTATOR – change value in a data field (void). (stuff)
public void setS(String s) { this.s=s; }
CONSTRUCTOR – used to create instances, must called. Don't return type or void. Same name as class
CLASS MEMBER – static. In dashed boxes
INSTANCE MEMBER – not static. Solid boxes
MEMBER – data field or method
DOT NOTATION – test.y (data field). test.t() (method)
TOSTRING – every instance has toString(). Can replace with custom text to print
ALIASES – a reference to a variable
REFERENCES – point to classes, including library classes
SERVICE METHOD – called on an object (dotnot) public
SUPPORT METHOD – called by other methods within the class eg. Test.t(). private
STATEMENTS – declarations, executables

t\ - tab
n\ - new line

/* */ = block comment
// = single line comment
/** */ = javadoc comment

Syntax error = compile error (code is not legal)
Run time error = code compiles, impossible to execute code, when you run the program
Semantic error = no error messages, code compiles and runs, doesn't do what you expected

Boolean – 1 bit
Byte – 8 bit
Short, Char – 16 bit
Int, Float – 32 bit
Long, Double – 64 bit
fractions/decimals = float, double
Int (whole numbers) = byte, short, int, long
true/false = boolean
letters/digits/symbols = char
Strings are instance objects (reference type)

```
Import java.util.Scanner;
Scanner sc = new Scanner(System.in);
System.out.println("some text here");
String s = sc.nextLine();
```

+=, -=, *=, /=, %=
I++ = evaluated to old then increments
++i = increments then evaluated to new

Math auto import with java.lang
Math.PI, Math.E, Math.sqrt(), Math.sin(), Math.max(), Math.pow(x,y)(x to the power of y)

widening ->
char, int, double
narrowing <-

double d = 1 (int conversion 1 to double 1.0)
double/int (int conversion to double. Result double)
int i = (int) 15.32 (cast double 15.32 to int 15)

Import javax.swing.*; - modern
Import java.awt.*; - old, builds on swing

Components = windows, frames, panels, buttons, text fields, scroll bars, check boxes, menus etc.

Java graphics are self contained (lightweight) and thus portable. Basic functions draw on the devices OS/windowing system (heavyweight).

All drawing code should be in a paintComponent
(public void paint(Graphics g)
g.drawLine(x,y,x,y), g.setColor(Color.red)
Graphics are drawn with an origin at the top left (x co-ord, y co-ord) YAXIS IS UPSIDE DOWN
drawRect(int x, int y, int width, int height), fillRect same
drawOval(Int x, int y, int width, int height), fillOval same
drawArc(int x, int y, int width, int height, int startAngle, int arcAngle), fillArc colours pie slices
drawString("Text", int x, int y)
More recent drawings done over top

```
Import javax.swing.JFrame; (or javax.swing.*;)
Public class Example0{
 Public static void main (String [] args){
   MyFrame obj1 = new MyFrame();
   obj1.setSize(int x, int y);
   obj1.setVisible(true);
   MyFrame obj2 = new MyFrame();
   obj2.setLocation(int x, int y);

Import javax.swing.*;    //need both for GUIs
Import javax.awt.*;
Public class MyFrame extends JFrame{
 Public void paint(Graphics g) {
     ALL DRAW THINGS GO HERE
```

length(); returns an int
indexOf(); returns an int
substring(start index, end index);
replace(old char, new char);
charAt(index);
equals("text"); use this instead of == for strings

```
import java.text.DecimalFormat;
DecimalFormat fmt = new DecimalFormat("0.###");

import java.text.NumberFormat;
NumberFormat fmt = new
NumberFormat.getCurrencyInstance()
or .getPercentInstance();
fmt.Format(object);
```

WRAPPER – wrap a primitive value into an object. Integer obj = new Integer(40). Byte, Short, Integer, Long, Float, Double, Character, Boolean, Void (cannot be instantiated). byteValue(), doubleValue() etc. return the value of this Integer as the corresponding primitive type. num = Interger.parseInt(str); (change int into string).

AUTOBOXING = automatic conversion between a primitive value and corresponding wrapper object. Reverse is UNBOXING

True || true is true
True || false is true
False || false is false
True && true is true
True && false is false
False && false is false

Method call
!, -(unary), +(unary)
New, type cast
*, /, %
+, -
<, <=, >, >=
==, !=
&&
||
=, +=, -=, /=, %=

```
if (condition){
 statements;
} else if (condition){
 statements;
} else {
 statements;
}

switch(n){
   case 1:
        System.out.println("something");
        Break;
   case 'a':
       Stuff here
    default:       (optional)
       Stuff
}
```
Have as many cases as you like, use breaks to end the switch after a case if you want

```
while(condition){      //only does if passes conditions
 statements;
}

do {                    //always does at least once
  statements;
} while (condition);

String [ ] words = {"The", "quick" etc };
for (String s : words){
   S.out.println(s);    //will print everything in 'words'
}

char c = 'a';

for(initialisation, condition, update)
 statement
Eg. for(int = 2, i <= 7, i++)

for(type var: collection){}
```

words[1] will access the word at that index in the list

private data fields enforce encapsulation. Methods should be public for service methods, private for support methods

```
p1 = new Point(7,4);
p2 = p1;
p3 = new Point(7,4);
```

p1 == p2 is true – aliases/references are equal
p1 == p3 is false – but p1.equals(p3) is true

this always refers to the current (instance) object.

```
Private int x,y;
 Public Point(x,y){
  this.x = x;
  this.y = y;
}
```

An array is a named collection of data of the same type and fixed size. Index starts from 0. Is an object. Can hold references to objects

int [] array1 = new int [5]; - 5 is the size of the array

```
for(int I = 0; i <array1.length; i++){
       x[i] = readInt("Enter an int for cell" + i);
}   //     how to fill an array with user entered ints
```

int [] x = {4,7};

a.length – length of an ARRAY called a
str.length(); - length of a STRING called str

int [] [] a = new int [3] [5]; - 3 down, 5 across
a[2][3] will access the element row 2 column 3
int [] [] a = {{2,3,4}{5,6,7}}

print a 2D array called b

```
for (int row = 0; row<b.length;row++){
 for (int col = 0; col < b[row].length; col++){
  System.out.print(b[row][col] + " ");
 }
 System.out.println();
}
```

or

```
for (double[] row : b){
 for (double x : row){
  System.out.print(x + " ");
 }
 System.out.println();
}
```

array a = b, c is the same nums as a/b
a == b is true
a == c is false

Arrays.equals(a,b) is true
Arrays.equals(a,c) is true

Main explained

```
public class Demo{
 public static void main (String [] args){
  System.out.println(args[0] + "***" + args[1]);
 }
}
```

GUI
COMPONENTS – objects drawn in the GUI eg buttons, also containers like panels and frames
EVENTS – objects that represent events eg button clicks
LISTENERS – objects that get sent events and have methods set up to process them
FRAME – a container that is drawn as a separate window. Heavyweight component. Several layers of panes, main one is content pane
PANEL – container can only exist as a subpart of another container (eg frame). Hold and organise others components. Lightweight component

Application class always has these 5 steps
1. Make frame/window
2. Define close behaviour
3. Add support panel
4. Auto size the frame (pack)
5. Make frame visible

Support classes always have these 3 steps
1. Define components
2. Add components to panel
3. Set panel details (eg size)

FlowLayout – left to right then top to bottom. Default
BorderLayout – north south east west center
BoxLayout – in a single row of column
GridLayout – within cells of a grid with standard sized rows and columns

add(b1, BorderLayout.SOUTH);

Containment Hierachies
1. Create components
2. Add components to panels
3. Add inner panels to the main/outer panel
4. Add the outer panel to the frame

INNER CLASS – declared inside another class. Only the outer can see/create instances of the inner. Members of outer are in scope for inner. Feature for safety and encapsulation
INTERFACE – like a restricted form of a class. Contain only abstract methods and static final data fields. Other classes may implement 1 or more interfaces. Enforce good design. Eg. implements WindowListener, ActionListener

event.getSource(), label.setText("text"), s.getText(), e.isSelected(),

CASTING – most general form of conversion in java. dollars = (int) money; result = (float) total/count;

Timer t = new Timer(1000, new MyListen());
t.start(); and t.stop(). 1000 is milleseconds, MyListen is the action listener method name

STREAM – is a source or destination of a sequence of data. Used as an abstraction for dealing uniformly with files and devices
TOKENS – text items separated by "white space" – blanks or tabs

Scanners use tokens. Scanner sc – new Scanner("axe bag cat");
System.out.println(sc.next()); etc for each token OR

```
while (sc.hasNext()){
    System.out.println(sc.next());
}
```
also nextInt(), nextDouble(), nextLine()

```
try{
  // code that might cause an exception
} catch (ExceptionClass variable){
  //warn the user to try to handle the exception
}
```

example exceptions. /0 = ArithmeticException, IOException (file read fail), InputMismatchException (unexpected data type)

outfile.close(); to close a file after you write it

reading from file – import java.io.*;

Sorting and Searching are main components of Comp Sci
BUBBLE SORT – swapping smaller neighbours with bigger ones
SELECTION SORT – keep putting the smallest thing at the start
MERGE SORT – split into two sub-lists, sort them, merge two lists
INSERTION SORT – take the next thing and put in order with past things
Selection sort has the same characteristics no matter which language or machine

EXTENDS – can extend almost any existing class to create/define a new class. A child class has its own members and also the non private members of its parent class

By default, every class extends java.lang.Object. toString is inherited from Object

```
public Person(String name, int age){
 this.name = name;
 this.age = age;
}
public Person(String name, int age, int height){
 this(name,age);
 this.height = height;
}
```
this() and super() have to be the first line in a method. Have one or the other but not both
SUPER = is a keyword referring to members of the current class/object inherited from the parent

CONSTRUCTOR CHAINING – every object is constructed by calling its own constructor, which calls a (maybe automatically) parent constructor, which calls a parent etc. up hierarchy to object constructor so objects are properly initialised

PACKAGES – group related classes together
CLASSES/VISIBILITY – set visibility of data fields. Abstraction. Encapsulation. Information hiding
METHODS/SCOPE – organise processing. Variables only within a method
BLOCKS {}/SCOPE – Organise processing. Can define local variables. In blocks within methods (scopes)
Packages and classes are for large programs/teams. Classes, methods, blocks are for small programs/individuals

Every class is part of a package. Eg java.lang. Classes you write/put in a package are in the default package

PACKAGES support OO concepts in java, involve some practical details about managing classes and files, define a "namespace" for the classes they contain. Classes in package must be stored in a single directory which should have the same name as the package

package java.lang; OR package com.sun.name;
import package.name;

CLASSPATH – tells compiler what directories to search
NAMESPACE – use fully qualified name eg java.util.Random or simple name (Random) if the class is in the same package or your source files import the other classes (java.util.*) or the other class is in java.lang
Importing a packages means we don't have to type out the full name every time we want to use the class

Visibility
PRIVATE – accessible within only this class
PACKAGE – accessible only to classes in the package. Doesn't particularly have a name because default. Promotes software reuse
PROTECTED – accessible to classes in this package and any classes extending this one
PUBLIC – accessible to all other classes (if class public)

ADT – abstract data type. All internal structor is hidden from the user. Can use classes as ADT by allowing access via public methods
INTERFACE – public methods of a class. Only interface needs to be understood. Everything else is hidden/safe (encapsulation/info hiding/ADT)

METHOD OVERLOADING – classes can have multiple methods with the same name as long as they have different signatures. Makes the structure of a program clear. If no matching signature it's a compiler method
METHOD OVERRIDING – when methods with the same signature exist in a parent class and a child class
BOTH PROMOTE COHESION
Overloading is shooting 3 arrows at once
Overriding is shooting 2 arrows taped together (long)

When you rewrite a method with the same signature in a child class the parent class becomes overridden and can only be accessed with super.methodName();
Same with SHADOWING (datafields with the same name)system.out.print (childClassName.variable); also

instanceOf – operator which returns true if an object is an instance of a class eq (s instanceOf Square)

Methods with the same name can exist in the same class with different signatures (overloading), a class and parent or other superclass with same signature (overriding), a class and its parent with different signatures (weak polymorphism)

Members are instance members (part of instance objects) unless the modifier STATIC specifies they are class members. Class data fields are shared by/have the same value for all instances of a class. Class methods can only access class data. Class members already exist as part of a class object when a program starts running. They are independent of any instance objects.

POLYMOPRHISM – ability of objects belonging to different types to respond to method calls of methods of the same name, each one according to an appropriate type-specific behaviour. Programmer and program don't have to know the exact type of the object in advance, so this behaviour can be implemented at run time (late binding or dynamic binding). Eg overloading, overriding

ABSTRACT CLASSES – used when you want to create instances of subclasses but NOT the abstract parent/ subclass class itself eg abstract Shapes because only create instances of Circles, Squares etc. Similar to interfaces in that you can treat subclasses as type Shape, ensure every method od type Shape will have certain methods and data fields. Different from interfaces in that you extend only one AC, but can implement many interfaces, and an AC can only have its normal methods (data fields don't need to be public static final)

```
Public abstract class Shape {
 public abstract void showArea();
```

```
}
public class Circle extends Shape{
 public void showArea(){
 stuff here no in the last class
 }}
```

Shapes [] can nclide circles, squares etc in AC

MULTIPLE INHERITANCE – eg extends x,y not allowed

FINAL – final class cannot be extended, overridden, or change data field once initialised and declared. Can be instantiated

ENUMERATED TYPES – are a way of creating a new type by enumerating (listing) every possible value

```
Enum Fruit {apple, orange, banana, cherry, grape}

Public static void main (String [] args){
 Fruit f = Fruit.apple;
}
```

f.ordinal() – returns ordinal position of f (from 0)
f.name() - returns value/name of f

```
for (Fruit fru : Fruit.values()){
 s.o.p("a nice " + fru);
}
```

GENERICS - programs can be written in terms of "to be specified later" types that are instantiated when needed for specific types (provided as parameters). Before explicit casts were needed now we can <v>

```
import java.util.ArrayList;

ArrayList a = new ArrayList();
a.add("Hello");
S.O.P(((String)a.get(0)).length());        //old
ArrayList<String> b = new ArrayList<String>();
b.add("Hello");
S.O.P(b.get(0).length());                  //new
```

COLLECTION API – includes ArrayList, copy(), min(), max(), fill(), reverse(), shuffle(), sort(), contains()

ARRAYLIST – can hold objects, don't need to declare size, can grow and shrink as required. Support generics. Can't store primitive types unless wrapper

In an ArrayList a.get(var/int), a.indexOf("name"), a.add("name"), a.add(index,"name"), a.remove(index)

If no type is specified, the ArrayList stores objects as type object. All below are allowed but it is risky. Depend on complier/IDE settings may get a warning

```
int [] x = {8,7,6}
ArrayList al = new ArrayList();
al.add("Hello");
al.add(new MyClass(42));
al.add(x);
```

to use objects in such a list must cast them to type
String s = (String)al.get(0);
((MyClass)al.get(1)).myMethod();

APPLETS – are not run directly like applications, but by and within another program called an applet context. Eg. Web browsers and applet viewers. They cannot access files on the machine. Run an applet by loading an html file into a browser. Browser calls the init method

GOOD DESIGN – packages, visibility, scope, inner classes, type checking, array bounds checking, references instead of pointers, garbage collection instead of memory allocation and deallocation

SIMULATION – is an attempt to model a real life situation on a computer so that it can be studied to see how the system works. By changing variables, predictions may be made about the behaviour of the system

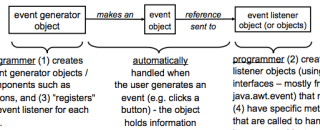
1. Solve the problem
2. Organise your program well
3. Keep your data safe
4. Write a big problem in small pieces
5. A working program may not be a good program
6. Anticipate errors
7. Start again instead of patching bad code
8. Clear code is better than clever code
9. Good formatting and commenting
10. Read the friendly manual (RTFM)

ENCAPSULATION – the characteristic of an object that means it protects and manages its own information
SEMANTICS – rules that define what a statement in a language means
INSTANTIATION – process of creating a new object
POLYMORPHISM – the ability to define an operation that has more than one meaning by having the operation dynamically bound to methods of various objects

Event model

3

The way a programming language handles events is called its "event model". Java's event model is comparatively simple and tidy!



programmer (1) creates event generator objects / components such as buttons, and (3) "registers" an event listener for each one.

automatically handled when the user generates an event (e.g. clicks a button) - the object holds information about the event.

programmer (2) creates listener objects (using interfaces – mostly from java.awt.event) that must (4) have specific methods that are called to handle input event objects.

Event model is the same for all kinds of events

6

Example generator	The user...	Event	Example listener interface
JButton	clicked a button	ActionEvent	ActionListener
JCheckBox	checked a box	ItemEvent	ItemListener
JScrollBar	moved a scroll bar	AdjustmentEvent	AdjustmentListener
JSlider	moved a slider bar	ChangeEvent	ChangeListener
JWindow	selected a window	FocusEvent	FocusListener
any Component	clicked/moved mouse	MouseEvent	MouseListener
JTextField	entered text	TextEvent	TextListener

For example, in a program handling events from a scroll bar, the four key steps of coding the event model might look like:

```
bar = new JScrollBar(); // Slide 3 Step (1)

bar.addAdjustmentListener( new MyListener() ); // Slide 3 Steps (2a), (3)

private class MyListener implements AdjustmentListener { // Slide 3 Step (2b)

public void adjustmentValueChanged(AdjustmentEvent event) { // Slide 3 Step (4)
```

```
import javax.swing.JFrame;

public class PizzaApp {

  public static void main (String[ ] args) {
    JFrame frame = new JFrame ( "Pizza Order" );
    frame.getContentPane().add( new PizzaOptionsPanel() );
    frame.setDefaultCloseOperation( JFrame.EXIT_ON_CLOSE );
    frame.pack();
    frame.setVisible( true );
  }

}
```

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class PizzaOptionsPanel extends JPanel {
  // the data fields are accessible by the constructor and by the inner classes (listeners)
  // declare the event generators (Event model Step 1)
  private JRadioButton [ ] size = { // create array of (refs to) JRadioButton objects
    new JRadioButton("Large"), new JRadioButton("Medium"), new JRadioButton("Small") };
  private JRadioButton [ ] base = { // create array of (refs to) JRadioButton objects
    new JRadioButton("Vegetarian"), new JRadioButton("Spicy"),
    new JRadioButton("Pepperoni") };
  private JCheckBox [ ] extras = { // create array of (refs to) JCheckBox objects
    new JCheckBox("Mushroom"), new JCheckBox("Onion"), new JCheckBox("Capsicum"),
    new JCheckBox("Olives") };
  // declare the output labels and the strings displayed in them
  private JLabel orderLabel = new JLabel("Order:  None");  // displays baseString,SizeString
  private JLabel extrasLabel = new JLabel("Extras:  None"); // displays extrasString
  private String baseString = "", sizeString = "", extrasString = "";  // strings to display

  /* constructor - creates the panels used for the GUI*/
  public PizzaOptionsPanel() {
    // this panel has layout positions NORTH WEST CENTER EAST SOUTH
    setLayout(new BorderLayout());
    // declare inner panels, one for each position
    JPanel headerPanel = new JPanel();  // to add at NORTH
    JPanel sizePanel = new JPanel();    // to add at WEST
    JPanel basePanel = new JPanel();    // to add at CENTER
    JPanel extrasPanel = new JPanel();  // to add at EAST
    JPanel orderPanel = new JPanel();   // to add at SOUTH

    // set up header panel - to add at position NORTH
    JLabel header = new JLabel();
    header.setFont(new Font("Helvetica", Font.BOLD, 18));
    header.setText("Create the pizza of your dreams");
    headerPanel.setPreferredSize( new Dimension(300,30) );
    headerPanel.setBackground( Color.red );
    headerPanel.add( header );

// set up size buttons and size panel - to add at position WEST
sizePanel.setPreferredSize( new Dimension(100,100) );
sizePanel.setLayout( new GridLayout(4,1) );    // holds 4 components (4 rows 1 column)
sizePanel.add( new JLabel("Select size:") );   // add label
ButtonListener bl = new ButtonListener();      // make listener (Event model Step 2a)
ButtonGroup sizeGroup = new ButtonGroup();  // buttons in a group are mutually exclusive
for (JRadioButton s : size) {
  s.addActionListener( bl );  // register listener for radio button  (Event model Step 3)
  sizeGroup.add( s );         // add button to group of mutually exclusive buttons
  sizePanel.add( s );         // add button to sizePanel
}

// set up base buttons and base panel - to add at position CENTER
basePanel.setPreferredSize( new Dimension(100,100) );
basePanel.setLayout( new GridLayout(4,1) );    // holds 4 components (4 rows 1 column)
basePanel.add( new JLabel("Select base:") );   // add label
ButtonGroup baseGroup = new ButtonGroup();  // buttons in a group are mutually exclusive
for (JRadioButton b : base) {
  b.addActionListener( bl ); // register listener for radio button   (Event model Step 3)
  baseGroup.add( b );        // add button to group of mutually exclusive buttons
  basePanel.add( b );        // add button to basePanel
}

// set up extras check boxes and extras panel - to add at position EAST
extrasPanel.setPreferredSize( new Dimension(100,100) );
extrasPanel.setLayout( new GridLayout(5,1) );   // holds 5 components (5 rows 1 column)
extrasPanel.add(new JLabel("Choose extras:"));  // add label
CheckListener cl = new CheckListener();         // make listener (Event model Step 2a)
for (JCheckBox e : extras) {
  e.addItemListener( cl );   // register listener for check box  (Event model Step 3)
  extrasPanel.add( e );      // add check box to extrasPanel
}

// set up order panel for text output in labels - to add at position SOUTH
orderPanel.setPreferredSize( new Dimension(300,50) );
orderPanel.setBackground( Color.green );
orderPanel.setLayout( new GridLayout(2,1) );   // holds 2 components (2 rows 1 column)
orderPanel.add( orderLabel );                  // add each label
orderPanel.add( extrasLabel );

// add inner panels to this panel at specified positions
add(headerPanel, BorderLayout.NORTH);
add(sizePanel, BorderLayout.WEST);

   add(basePanel, BorderLayout.CENTER);
   add(extrasPanel, BorderLayout.EAST);
   add(orderPanel, BorderLayout.SOUTH);
 } // end of constructor


 /* this inner class is the listener for all JRadioButtons (Event model Step 2b)*/
 private class ButtonListener implements ActionListener {
   public void actionPerformed (ActionEvent event) { // Event model Step 4
     Object source = event.getSource();   // record source of event (which radio button)
     for(JRadioButton s : size) {         // check all size buttons
       if (s == source) sizeString = s.getText(); // get text from source button and set sizeString
     }
     for(JRadioButton b : base) {         // check all base buttons
       if (b == source) baseString = b.getText(); // get text from source button and set baseString
     }
     // now set label text
     orderLabel.setText("Order: " + sizeString + " " + baseString);
   } // end of method
 } // end of ButtonListener class


/* this inner class is the listener for all JCheckBoxes   (Event model Step 2b) */
 private class CheckListener implements ItemListener {
   public void itemStateChanged (ItemEvent event) {  // Event model Step 4
     extrasString = "";  // must start from an empty string and check each box
     // check boxes have a useful boolean isSelected method!
     for(JCheckBox e : extras) {
       // if this box is selected, get its text and add to the output string
       if (e.isSelected() ) extrasString = extrasString + e.getText() + " ";
     }
     // now set label text
     extrasLabel.setText("Extras: " + extrasString);
   } // end of method
 } // end of ButtonListener class

} // end of PizzaOptionsPanel class
```

Example 2

10

Assume a file where each line is a name, id number, and lab marks (max of 50 lines). This example will process every line, reading the id numbers into an array.

File "data2.txt".

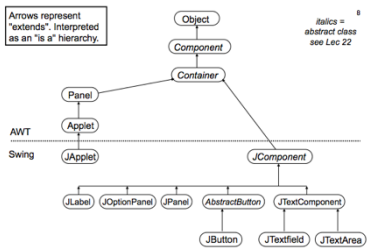
```
Anne 34792 1 1 1 1 0
Barry 42424 1 1 0 0 1
Craig 84421 0 0 0 1 1
etc…
```

- create the array (max of 50)
- creates a Scanner for the file
- checks to see if the file has a next token, and if it does…
- creates a Scanner for the line of the file (treated as one string)
- throws away first token (name)
- reads the second token as an int and stores it in the array.

```
// imports as for Example 1
int [ ] ids = new int [50];  // declare array
try {
  Scanner sc = new Scanner(new File("data2.txt") );
  int i = 0;  // initialise index for array
  while ( sc.hasNext() ) {
    Scanner line = new Scanner( sc.nextLine() );
    line.next();  // read / discard first token
    ids[ i ] = line.nextInt();  // read next token as int
    i++;  // increment array index
  }
} catch ( IOException ioex ) {
  // some problem with the file (not found?)
  System.out.println("File problem!: " + ioex );
} catch ( InputMismatchException imex ) {
  // a data value was not as expected
  System.out.println("Unexpected data!: " + imex );
}
```

ids → [0] 34792, [1] 42424, [2] 84421, etc…

8

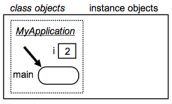


16

Application

```
public class MyApplication {
  private static int i = 2;
  public static void main(String [ ] args) {
    // program starts running here
    System.out.println("i is: " + i);
    // output written to Java console
  }
}
```

class objects — instance objects



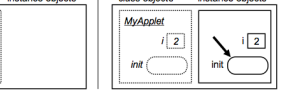
Applet

```
import javax.swing.JApplet;

public class MyApplet extends JApplet {
  private int i = 2;
  public void init( ) {
    // program starts running here
    System.out.println("i is: " + i);
    // output written to browser log file
  }
}
```

log is browser dependent - text output is not usual for an applet!

class objects — instance objects




```
package shapes;
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;
import java.util.ArrayList;

/** ShapePanel.java
  * Amanda Veldman September 2016
  * creates a frame and adds the control panel and the drawing panel */

public class ShapePanel extends JPanel{

  ArrayList<Shape> shapes = new ArrayList<Shape>();
  private DrawingPanel drawPanel;
  private JPanel controlPanel;
  private JButton circle;
  private JButton square;
  private JButton smiley;
  private JButton oval;
  private JButton swirl;
  private JButton start;
  private JButton stop;
  private JButton remove;
  private JTextField showNum;
  private JLabel removeLabel;
  private int count;
  Timer timer;
  private final int DELAY = 5;                  // data field declerations

  JButton [] buttons = new JButton[7];

  /*creates and displays the traffic light program frame */
  public static void main(String [] args){

    JFrame frame = new JFrame("Lab 21");
    frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

    frame.getContentPane().add(new ShapePanel());

    frame.pack();
    frame.setVisible(true);
  }

  /** Constructor */
  public ShapePanel(){

    controlPanel = new JPanel();
    circle = new JButton("Circle");
    square = new JButton("Square");
    smiley = new JButton("Smiley");
    oval = new JButton("Oval");
    swirl = new JButton("Swirl");
    start = new JButton("Start");
    stop = new JButton("Stop");
    remove = new JButton("Remove");
    showNum = new JTextField(2);
    removeLabel = new JLabel("Remove which?");
    drawPanel = new DrawingPanel();          // creates all components

    JButton [] buttons = {circle, square, smiley, oval, swirl, start, stop, remove};

    ButtonListener listener = new ButtonListener();
    for (int i = 0; i < buttons.length; i++){
      buttons[i].addActionListener(listener);
      controlPanel.add(buttons[i]);                  // adds listeners to the buttons
    }

    timer = new Timer(DELAY, listener);

    controlPanel.add(removeLabel);                   // add components to the control panel
    controlPanel.add(showNum);

    add(controlPanel);
    add(drawPanel);                                  // adds panels to the frame

    controlPanel.setPreferredSize(new Dimension(100,400));
  }
/** inner priavte class
  * listens for ActionEvents and creates shapes in the shapes array */
private class ButtonListener implements ActionListener{

  public void actionPerformed (ActionEvent event){
    if (event.getSource() == timer){
      for(Shape s:shapes){
        s.move();
      }
    } else {
      JButton button = (JButton) event.getSource();
      if (button.getText().equals("Circle")){
        shapes.add(new Circle());
        count ++;
      } else if (button.getText().equals("Square")){
        shapes.add(new Square());
        count ++;
      } else if (button.getText().equals("Smiley")){
        shapes.add(new Smiley());
        count ++;
      } else if (button.getText().equals("Oval")){
        shapes.add(new Oval());
        count ++;
      } else if (button.getText().equals("Swirl")){
        shapes.add(new Swirl());
        count ++;
      } else if (button.getText().equals("Start")){
        timer.start();
      } else if (button.getText().equals("Stop")){
        timer.stop();
      } else if (button.getText().equals("Remove")){
        try{
        int in = Integer.parseInt(showNum.getText());
        shapes.remove(in);
        } catch (NumberFormatException z){
          System.out.println("No more shapes left to remove");
        } catch (IndexOutOfBoundsException b){
          System.out.println("There arent that many shapes");
        }
      }
      if (shapes.size() == 0){
        showNum.setText("");
      } else {
        showNum.setText(Integer.toString(shapes.size()-1));
      }
      drawPanel.repaint();
    }
  }

/** inner priavte class
  * drawing panel draws the shapes in the array shapes*/
private class DrawingPanel extends JPanel{

  public DrawingPanel(){
    setPreferredSize(new Dimension(400,400));
    setBackground(Color.pink);
  }

  public void paintComponent(Graphics g){
    super.paintComponent(g);
    for(Shape s:shapes){
      s.display(g);
      s.showIndex(g, shapes.indexOf(s));
    }
  }
}
```

```
package shapes;
import java.awt.*;
import javax.swing.*;
import java.util.Random;

/** Shape.java
  * Amanda Veldman September 2016
  * creates circles of random sizes and colours */

public abstract class Shape{

  int x;
  int y;
  int width;
  int height;
  Color colour;
  int moveX = 1;
  int moveY = 1;
  int change = 1;

  /** @return random number between the range given */
  public int randomRange(int lo, int hi){
    Random generator = new Random();
    int rand;

    rand = generator.nextInt(hi) + lo;
    return rand;
  }

  /** counstructor method
    * sets random height, width and colour */
  public Shape(){
    Random generator = new Random();
    width = generator.nextInt(20) + 10;
    height = width;

    x = generator.nextInt(400-width);
    y = generator.nextInt(400-height);

    colour = new Color(generator.nextInt(255), generator.nextInt(255), generator.nextInt(2
  }

  /** alters x, y, and colour to create the animation */
  public void move(){
    Random generator = new Random();
    if (change % 2 == 0){
      colour = new Color(generator.nextInt(255), generator.nextInt(255), generator.nextInt
      change++;
    } else {
      change ++;
    }                                  // colour change every 2 increments of change

    if (x >= 400-width){
      moveX = -moveX;
    } else if (x <= 0 ) {
      moveX = (-1)*(moveX);
    }                                  // increments moveX
    if (y >= 400-height){
      moveY = -moveY;
    } else if (y <= 0 ) {
      moveY = (-1)*(moveY);
    }                                  // increments moveY
    if (width > 15){
      y += moveY;
    } else {
      x += moveX;
    }                                  // wider than 15 pixels move up and down, else sideways
  }

  /** method to show the index of each shape */
  public void showIndex(Graphics g, int i){
    String z = Integer.toString(i);
    g.setColor(Color.black);
    g.drawString(z, x, y);
  }

   /** draws a random val based on the randomly set data fields */
   public abstract void display(Graphics g);

}
```

```
package shapes;
import java.awt.*;
import java.util.Random;

/** Smiley.java
  * Amanda Veldman September 2016
  * creates smileys. extends the shape abstract class */

public class Smiley extends Shape{

  /** constructor
    * sets both height and width to 30 pixels always */
  public Smiley(){
    height = 30;
    width = 30;
  }

  public void display(Graphics g){
    g.setColor(colour);
    g.fillOval(x, y, width, height);
    g.setColor(Color.black);
    g.drawOval(x+7, y+8, 4, 4);
    g.drawOval(x+20, y+8, 4, 4);
    g.drawArc(x+8, y+10, 15, 13, 190, 160);
  }
}
```